

Analyzing Patient Health Data

Nutrition Measurements Dashboard



• Patient Insights



• Nutrition Analysis



• Data-Backed Decisions



• Empowering better health Outcomes



Intermountain
Healthcare

Project Overview



Nutrition Measurements Dashboard

Analyzing Patient Health Data

As an Apprentice Data Analyst at Intermountain Healthcare, I contributed to the development of a Nutrition Measurements Dashboard for Inpatient Rehabilitation Services.

PROJECT GOALS

Identify Outliers

Detect unusual changes in patient nutrition measurements.

Support Clinical Decisions

Provide clinicians with clear, timely insights to support patient care.

Monitor Patient Outcomes

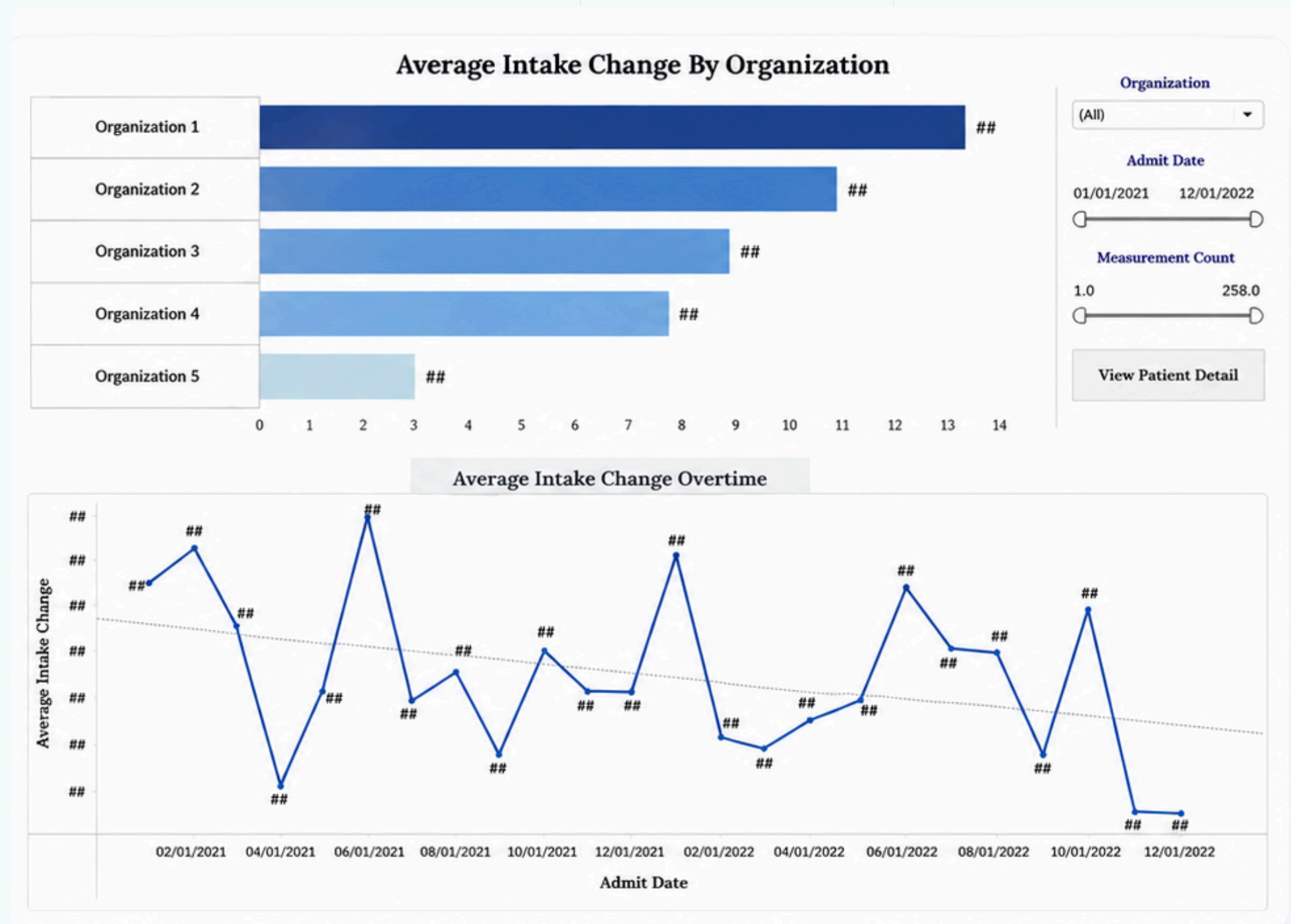
Track nutrition trends over time to help teams evaluate patient progress.

Nutrition Measurements Dashboard

Interactive Tableau dashboard for monitoring patient nutrition trends and identifying unusual changes.

DASHBOARD FEATURES

- Organization-level comparison
- Nutrition trends over time
- Patient-level filtering
- Outlier identification



Data Preparation & Analysis

How I Built the Dashboard

Preparing and transforming patient nutrition data for analysis.

Prepare

Cleaned and validated patient nutrition data using Oracle SQL.

Integrate

Used SQL JOINS to combine relevant patient and encounter data.

Transform

Created Tableau calculated fields to standardize nutrition measurements.

Analyze

Calculated intake changes to identify meaningful patient-level trends.



Oracle SQL — Data preparation & integration

```
SELECT  MESURE_VAL, NEXT ORDER FROM FIRST ORDER BY CD.REPORTED_DTS) AS First_Intake
        COUNT(MESURE_VAL, NEXT ORDER FROM LAST ORDER BY CD.REPORTED_DTS) AS Last_Intake
        COUNT(DISTINCT MRN) AS Total
        SUM(CD.FIRST_INTAKE_DTS IS NULL) Intake_DTS
        SUM(CD.REPORTED_DTS IS NULL) Last_Intake_DTS

FROM CD
LEFT JOIN patient_encounter pe ON pe.person_id = CD.person_id
LEFT JOIN patient_location pl ON pl.person_id = CD.person_id
LEFT JOIN location_patient_care ipc ON ipc.location_id = pl.location_id
WHERE CD.location_event_id = pe.event_id OR CD.event_id = pl.event_id;
```


From Data to Clinical Insights

Turning prepared data into actionable dashboard metrics.

Improved Visibility

Made patient nutrition trends easier to monitor.

Faster Identification

Helped surface unusual intake changes for further review.

Decision Support

Provided clinicians with accessible data to support patient monitoring.



Tableau — Calculation Development

FIRST INTAKE %



```
IF [First Intake] = '1-25%' THEN 25
ELSEIF [First Intake] = '26-50%' THEN 50
ELSEIF [First Intake] = '51-75%' THEN 75
ELSEIF [First Intake] = '76-100%' THEN 100
ELSEIF [First Intake] = '90-100%' (Disorder eating protocol') THEN 100
ELSEIF [First Intake] = 'None' THEN 0
ELSEIF [First Intake] = 'NPO' THEN 0
END
```

Interactive Tableau Dashboard

Weight Loss & Nutrition Intake Analytics

Dynamic Visualizations

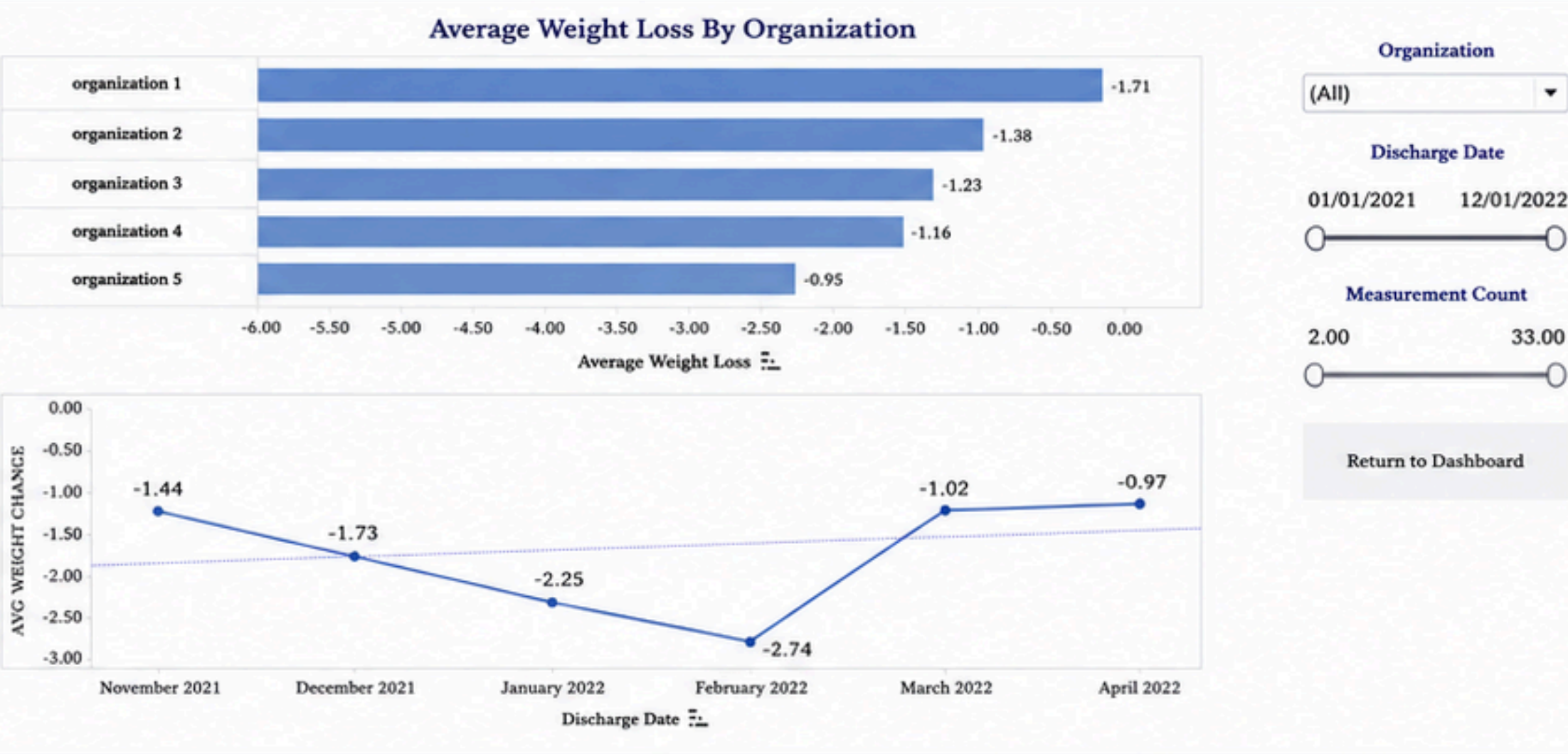
Built interactive views to track weight change and nutrition intake trends across organizations and over time.

Enhanced Usability

Added intuitive filters and consistent navigation to make the dashboard easier to explore.

Patient-Level Insights

Created detailed patient views and outlier indicators to help identify unusual changes for further review.



Patient Weight Loss Details

Fin Number	Patient	Admit Date	Discharge Date	Organization	Order Status	Measurement Count	First Weight	Last Weight	Weight Change
fin # 1	patient 1	1/10/2021	07/30/2021	Intermountain Medical Center	OUTLIER	5.0	187.7	80.9	-107.70
fin # 2	patient 2	1/21/2022	02/04/2022	Intermountain Medical Center	OUTLIER	2.0	194.9	87.2	-107.60
fin # 3	patient 3	8/17/2022	08/20/2022	Mckay-Dee Hospital	OUTLIER	18.0	175.8	86.0	-106.80
fin # 4	patient 4	2/28/2021	03/16/2021	Mckay-Dee Hospital	OUTLIER	12.0	217.7	114.1	-103.60

Fin Number

Patient

(All) ▼

Discharge Date

01/05/2021 12/01/2022

Organization

(All) ▼

Order Status

(All) ▼

[Return to Dashboard](#)

Project Summary

Nutrition Measurements Dashboard



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MY CONTRIBUTIONS

1

Prepared the Data

Cleaned, validated, and integrated nutrition data using Oracle SQL.

2

Developed the Analysis

Created SQL and Tableau calculations to measure nutrition and weight changes.

3

Built the Dashboard

Developed an interactive Tableau dashboard with organization- and patient-level filters.

4

Tested & Collaborated

Refined the dashboard with clinical and operational stakeholders to improve usability and reporting accuracy.

PROJECT OUTCOME

Delivered an interactive dashboard that gave stakeholders greater visibility into patient nutrition trends and helped identify unusual changes for further clinical review.

PROJECT OUTCOME

The dashboard transformed patient nutrition data into an interactive tool for monitoring trends, identifying outliers, and supporting clinical review.

By combining Oracle SQL and Tableau, the project made complex patient-level data easier to explore and interpret.

NEXT STEP

Continue refining the dashboard through user feedback and expanded analytical functionality.

thank you!

